Design of a RC beam

$b := 300\ mm$ - beam width

$h := 450\ mm$ - beam height

$l := 6000\ mm$ - span length

$\phi := 20\ mm$ - reinf. diameter - T

$\phi' := 14\ mm$ - reinf. diameter - C

$link := 8\ mm$ - diameter of links

$c_{nom} := 35\ mm$ - concrete cover

$g_k := 30.2\ \frac{kN}{m}$ - permanent load

$q_k := 11.5\ \frac{kN}{m}$ - variable load

$M_{Ed} := 184\ kN \cdot m$ - design bending moment

$V_{Ed} := 165\ kN$ - design shear force

$f_{ck} := 30\ MPa$ - for concrete C30/37

$\gamma_c := 1.5$ - partial factor for concrete

$f_{yk} := 500\ MPa$ - characteristic yield strength

$\gamma_s := 1.15$ - partial factor for reinforcing steel

$f_{yd} := \frac{f_{yk}}{\gamma_s}$ - design yield strength

$f_{ywk} := 500\ MPa$ - characteristic yield strength of shear reinforcement

$f_{ywd} := \frac{f_{ywk}}{\gamma_s}$ - design yield strength of shear reinforcement

$f_{ctm} := 0.3 \cdot \sqrt[3]{f_{ck}^{\ 2}}\ MPa$ - mean tensile strength

$\delta := 0.85$ - redistribution ratio

$d_g := 20\ mm$ - maximum agregate size

Flexural design:

$d := h - c_{nom} - link - \frac{\phi}{2} = 397\ mm$ - effective depth

$d_2 := c_{nom} + link + \frac{\phi'}{2} = 50\ mm$

$K := \frac{M_{Ed}}{b \cdot d^2 \cdot f_{ck}} = 0.1297$

$K' := 0.6\ \delta - 0.18\ \delta^2 - 0.21 = 0.17$

$$z := \begin{cases} \text{if } K \le K' \\ \quad \min\left(\frac{d}{2}\left(1 + \sqrt{1 - 3.53\ K}\right), 0.95\ d\right) \\ \text{else} \\ \quad \frac{d}{2}\left(1 + \sqrt{1 - 3.53\ K'}\right) \end{cases}$$

$z = 345\ mm$

$x := 2.5\ (d - z) = 131\ mm$

$f_{sc} := \min\left(\frac{700\ MPa \cdot (x - d_2)}{d_2}, \frac{f_{yk}}{1.15}\right) = 435\ MPa$

$$A'_{s.req} := \begin{cases} \text{if } K \le K' \\ \quad 0\ mm^2 \\ \text{else} \\ \quad \frac{(K - K') \cdot f_{ck} \cdot b \cdot d^2}{f_{sc} \cdot (d - d_2)} \end{cases}$$

$A'_{s.req} = 0\ mm^2$ - area of required comp. reinforcement

For $N'_o := 0$ $\phi' := 0\ mm$

$A'_{s.prov} := N'_o \cdot \pi \cdot \frac{\phi'^2}{4} = 0\ mm^2$ - area of provided reinforcement

$$A_{s.req} := \begin{cases} \text{if } K \le K' \\ \quad \dfrac{M_{Ed}}{f_{yd} \cdot z} \\ \text{else} \\ \quad \dfrac{K' \cdot f_{ck} \cdot b \cdot d^2}{f_{yd} \cdot z} + A'_{s.req} \cdot \dfrac{f_{sc}}{f_{yd}} \end{cases}$$

$A_{s.req} = 1228\ mm^2$ - area of required tension reinf.

For $N_o := 4$ $\phi := 20\ mm$ $A_{s.prov} := N_o \cdot \pi \cdot \dfrac{\phi^2}{4} = 1257\ mm^2$ - area of provided reinforcement

$M_{Rd} := A_{s.prov} \cdot z \cdot f_{yd} = 188\ kN \cdot m$ - bending cpacity > $M_{Ed} = 184\ kN \cdot m$

$$A_{s.min} := \max\left(\frac{0.26\ f_{ctm} \cdot b \cdot d}{f_{yk}}, 0.0013\ b \cdot d\right) = 179\ mm^2$$

$A_{s.max} := 0.04\ b \cdot h = 5400\ mm^2$

Check spacing:

$$Spacing := \frac{\left(b - 2\ c_{nom} - 2\ link - N_o \cdot \phi\right)}{N_o - 1} = 45\ mm$$

$a := \max\left(20\ mm, \phi, \left(d_g + 5\ mm\right)\right) = 25\ mm$ - min clear distance between bars

Deflection:

$\rho := \dfrac{A_{s.req}}{b \cdot d} = 10.31 \cdot 10^{-3}$ $\rho_0 := 10^{-3} \sqrt{\dfrac{f_{ck}}{MPa}} = 5.48 \cdot 10^{-3}$

$\sigma_s := \dfrac{f_{yk}}{\gamma_s} \dfrac{A_{s.req}}{A_{s.prov}} \dfrac{g_k + q_k}{1.35\ g_k + 1.5\ q_k} = 305\ MPa$ - reinforcement stress under characteristic load

$K := 1.0$ - factor dependent from the structural system - table 7.4N (BS EN 1992-1-1)

$\rho' := \dfrac{A'_{s.req}}{b \cdot d} = 0$ - required compression reinforcement ratio

$$l_{to}d := \begin{cases} \text{if } \rho \le \rho_0 \\ \quad K \cdot \left(11 + 1.5 \sqrt{\dfrac{f_{ck}}{MPa}} \dfrac{\rho_0}{\rho} + 3.2 \sqrt{\dfrac{f_{ck}}{MPa}} \left(\dfrac{\rho_0}{\rho} - 1\right)^{\frac{3}{2}}\right) \cdot \dfrac{310\ MPa}{\sigma_s} \\ \text{else} \\ \quad \left(11 + 1.5 \sqrt{\dfrac{f_{ck}}{MPa}} \dfrac{\rho_0}{\rho - \rho'} + \dfrac{1}{12} \sqrt{\dfrac{f_{ck}}{MPa}} \sqrt{\dfrac{\rho'}{\rho}}\right) \cdot \dfrac{310\ MPa}{\sigma_s} \end{cases}$$

- value of the limiting span/depth ratio

$l_{to}d = 15.6$ > $\dfrac{l}{d} = 15$ - actual span/depth ratio, satisfactory

Shear design:

$C_{Rd.c} := \frac{0.18\ MPa}{\gamma_c} = 0.12\ MPa$ $\quad \sigma_{cp} = 0\ MPa$ $\quad k_1 := 0.15$

$k := min\left(1 + \sqrt{\frac{200\ mm}{d}}, 2\right) = 1.71$ $\quad \alpha_{cw} := 1$ - for non prestressed structures (compression chord coeffitient)

$\rho_1 := min\left(\frac{A_{s.prov}}{b \cdot d}, 0.02\right) = 0.011$ $\quad f_{cwd} := \frac{\alpha_{cw} \cdot f_{ck}}{\gamma_c} = 20\ MPa$

$\nu_{min} := 0.035\ MPa \cdot k^{\frac{3}{2}} \cdot \sqrt{\frac{f_{ck}}{MPa}} = 0.429\ MPa$ $\quad \nu := 0.6\left(1 - \frac{f_{ck}}{250\ MPa}\right) = 0.528$ - strength reduction factor

$V_{Rd.c} := \left(C_{Rd.c} \cdot k \cdot \left(100\ \rho_1 \cdot \frac{f_{ck}}{MPa}\right)^{\frac{1}{3}} + k_1 \cdot \sigma_{cp}\right) b \cdot d = 77.3\ kN$ - shear capacity without links

$V_{Rd.c} = 77.3\ kN$ > $\nu_{min} \cdot b \cdot d = 51\ kN$

$V_{Rd.c} = 77.3\ kN$ < $V_{Ed} = 165\ kN$, hence shear reinforcement must be provided

$V_{Rd.max.21.8} := \nu \cdot f_{cd} \cdot b \cdot z \cdot \cos(21.8\ deg) \cdot \sin(21.8\ deg) = 376\ kN$

$V_{Rd.max.45} := \nu \cdot f_{cd} \cdot b \cdot z \cdot \cos(45\ deg) \cdot \sin(45\ deg) = 546\ kN$ - angle of compression shear strut for maximum shear

$$\theta := \begin{cases} \text{if } V_{Ed} < V_{Rd.max.21.8} \\ \quad 21.8\ deg \\ \text{else if } V_{Rd.max.21.8} < V_{Ed} < V_{Rd.max.45} \\ \quad 0.5 \cdot \operatorname{asin}\left(\frac{3\ V_{Ed}}{\nu \cdot f_{ck} \cdot b \cdot z}\right) \end{cases}$$

$\theta = 21.8\ deg$ - angle of concrete compression strut ($21.8 < \theta \leq 45°$)

$A_{sw.req} := \frac{V_{Ed}}{0.78 \cdot d \cdot f_{yk} \cdot \cot(\theta)} = 0.426\ \frac{mm^2}{mm}$ - required transverse reinforcement

For $N_o := 2$ shear legs $\quad \phi := 8\ mm$ $\quad A_{sw.prov} := N_o \cdot \pi \cdot \frac{\phi^2}{4} = 101\ mm^2$

$s_w := \frac{A_{sw.prov}}{A_{sw.req}} = 236\ mm$ < $s_{w.max} := min(0.75\ d, 600\ mm) = 298\ mm$

$s_w := 200\ mm$ - adopted distance between links

$\rho_w := \frac{A_{sw.prov}}{b \cdot s_w} = 0.17\%$ > $\rho_{w.min} := \frac{0.08\ MPa \cdot \sqrt{\frac{f_{ck}}{MPa}}}{f_{yk}} = 0.09\%$ - min % of shear reinf.

$V_{Rd.s} := \frac{A_{sw.prov}}{s_w} \cdot z \cdot f_{ywd} \cdot \cot(\theta) = 188\ kN$ - shear capacity > $V_{Ed} = 165\ kN$

Where $V_{Ed} > V_{Rd.c}$, shear reinf. must be provided and min shear reinf. for the rest of the beam.